

Lab 11

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In [ ]: import numpy as np
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In [ ]: X = np.array([[0,0],[0,1],[1,0],[1,1]])
Y = np.array([[0],[1],[1],[1]])
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In [ ]: weights = np.array([0.5,0.7])
bias = 0.9
lr = 0.1
epochs = 5000
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In [ ]: def step_fun(n):
    if n >= 0:
        return 1
    return 0
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In [ ]: for epoch in range(epochs):
    print(f"\nEPOCH = {epoch+1}")

    total_mistakes = 0

    for i in range(4):
        x = X[i]
        y = Y[i]
        z = np.dot(x,weights) + bias
        pred = step_fun(z)
        print(f"{x}--{y}--{z}--{pred}")

        error = y-pred

        if error != 0:
            weights = weights + (lr * error * x)
            bias = bias + (lr * error * 1)
            total_mistakes += 1
        else:
            print(f"Corrected on {x}")
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In [ ]: def My_Or(a,b):
    z = np.dot((a*weight[0]+b*weight[1]))+bias
    pred = stepfunction(z)
    return pred
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In [ ]:
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